

1. Introduction

In today's fiercely competitive retail industry, baked goods - characterised as high-frequency, fast-moving consumer goods - often exhibit significant daily demand fluctuations, especially in the baking departments of cake shops or large supermarkets. These fluctuations may also be influenced by external factors such as weather [1]. For example, temperature changes may alter people's cravings for sweets or high calorie foods, while rainfall can greatly reduce customers' willingness to go shopping, directly affecting the sales of baked goods. On the other hand, holidays and promotional activities largely determine the highs and lows of baked goods sales [2]. Excessive inventory can lead to product expiration, waste, and decreased profits, while insufficient inventory can result in unmet customer needs, thereby affecting sales revenue and customer satisfaction [3]. Therefore, establishing a model that can accurately predict daily baking product sales is crucial for optimising inventory management and production planning [4].

In recent years, there has been a significant shift in sales forecasting models in the retail industry, shifting from relying on traditional time series analysis methods such as ARIMA and SARIMA to more complex and refined machine learning and deep learning models such as Random Forest, XGBoost, and LSTM [5]. During this process, many researchers and practitioners actively explored various improvement strategies for time series methods. These new strategies aim to further improve prediction accuracy by introducing new algorithms or optimising existing model structures to handle unstable and complex demand patterns in retail scenarios [6].

Relying solely on historical sales data often fails to accurately capture the various influencing factors behind sales volume. Therefore, incorporating multidimensional external variables such as weather data, holiday information, customer behaviour patterns, etc. into the prediction model can significantly improve the accuracy and robustness of sales forecasting [7]. In time series forecasting research, Hyndman and Athanasopoulos emphasised the importance of identifying seasonal and periodic patterns, and pointed out that if these patterns are captured correctly, the model can achieve more accurate predictions [8]. Meanwhile, Box et al. proposed the Box Jenkins method (ARIMA model family), which provides a systematic modelling and diagnostic process for data exhibiting strong stationarity or seasonality, laying a solid theoretical and practical foundation for many beginners and practitioners [9]. In the field of machine learning, the random forest model introduced by Breiman can reduce the risk of overfitting and improve prediction accuracy by integrating multiple decision trees [10].

The core goal of this "Daily Sales Forecast for Store Baking Products" project is to establish a high-precision forecasting model using historical supermarket sales data and multiple external data sources (weather, temperature, holidays, promotional activities, etc.) to provide data-driven decision support for inventory and production management. The model will determine which variables have an impact on daily average sales, which variables should be included in each model, which model should be used, and to what extent we can make the model accurate in predicting daily sales. According to the model structure, a plan will be proposed to reduce overproduction and inventory surplus, which will enable enterprises to effectively reduce costs; Developing this method can help maximise full price sales, reduce waste, lower costs, and improve customer satisfaction. Given that different models have different advantages in processing time series data, this study plans to adopt a hybrid modelling approach, combining time series models with machine learning or deep learning models to obtain more robust prediction results [6] [9]. In this report, we first perform data cleaning on data from 33 stores to reduce noise and focus on data with clear features, particularly addressing the relationship between inventory and demand. Then, we proposed some hypotheses that rainfall, day of the week, and season have a significant impact on sales volume. Based on these assumptions, we used feature engineering to confirm that these variables do impact sales. Next, we compared several popular models to see how they predict sales volume and selected the model that has been proven to be the most effective. Finally, we summarised the prediction results of the model, analysed and evaluated the generated data, and obtained decision-making insights.

2. Data preprocessing

2.1. Data Collection and Integration

In this project, the data is mainly divided into the following two parts:

1. Sales data:

- a) The supermarket baking department provides historical sales records of multiple products, including product names, store names, sales dates, sales quantities, total sales prices, etc.
- b) The time frame of this dataset exceeds 3 years, and the corresponding seasons and weeks are provided, which can effectively capture seasonal and periodic features and make periodic predictions [1].

2. External data:

- a) Weather data: sourced from the marketing department of the enterprise and data obtained from the meteorological bureau. Weather information includes meteorological variables such as perceived temperature, precipitation, and cloud cover.
- b) Promotional activity information: sourced from the marketing department of the enterprise, including discount and other promotional types and promotional efforts.

By integrating this information, we can gain a more comprehensive understanding of the potential factors that impact changes in sales volume.

2.2. Data cleaning

1. Handling of missing values

There may be missing records in the dataset, such as sales data loss caused by incomplete promotional information or system failures [4]. For abnormal data, if there are small-scale anomalies, we will fill them with the mean, and if there are large-scale missing data, we will delete the corresponding data.

2. Outlier detection and processing management

In sales or price data, the presence of abnormal peaks or abnormally low values often affects the effectiveness of model training [10]. Visual methods can be used, such as using boxplots for preliminary inspection and observation of abnormal data [11], and deleting abnormal data values.

2.3. Noise elimination and smoothness testing

1. White noise elimination

White noise refers to completely random and irregular data. We used tools such as Python and the statsmodels library to assist with calculations, and based on the reported results, we eliminated the impact of white noise in the data.

2. Smoothness detection

The stationarity test is to check whether our data has "stability". In order to detect smoothness, this article performed a "differencing" process on the data. It focuses on the amount of change, not the specific quantity. Through differencing, some smoothness can be observed, and Python's basic library can be used to improve data smoothness.

2.4. Data selection and splitting

1. Differentiation between training and testing sets

Due to the temporal nature of time series data, it is usually divided in chronological order [5]. In this project, after sorting the data by time, the most recent data (1-2 months) is kept as the test set to test the model's generalisation ability in actual business scenarios [8].

2. Target dataset filtering

Based on the objective to be explored, we observed that single roll has obvious specificity and calculated the average sales of stores. We selected four stores with smaller data scales as the target dataset for analysis. During the analysis, we found that 8040 had some missing products, so we also removed this store.

3. Model Choice and Justification

For sales forecasting, we selected a diverse set of models to address the unique challenges of analysing time series and structured data. Each model was selected based on its ability to account for certain aspects of sales dynamics, including time dependence, seasonality, feature interactions, and interpretability.

3.1. Long-term memory and short-term memory (LSTM)

Reasons for choice:

- Excellent for learning long-term temporal dependencies in sequential data [12].
- Ability to handle non-linear relationships and adapt to irregular patterns (e.g. fluctuations in demand due to seasonal shifts).
- Robust against noise and missing values thanks to managed memory mechanism.

For sales forecasts:

Long short-term memory (LSTM) networks are well suited for modelling complex sales behaviours at multiple scales where models exhibit non-linear and/or noisy temporal patterns.

3.2. Gradient Boosting (LightGBM)

Reasons for choice:

- Structured tabular data optimised for fast and efficient learning [13].
- Ability to process mixed data types (numeric, categorical) and auto-encode interactions between features.

For sales forecasts:

LightGBM uses auxiliary data and time series features to create complex forecasts while preserving interpretability.

3.3. Random Forest

Reasons for choice:

- Ensemble methods that reduce overfitting by clustering features and randomizing [10].
- Nonparametric models account for skewed distributions and outliers that are common in sales data.
- Provides reliable estimates of feature importance for hypothesis testing (e.g. impact of regional sales patterns).

For sales forecasts:

Random forests provide a robust benchmark for testing complex models and are particularly useful when analysing data sets with heterogeneous features.

3.4. SARIMA

Rationale for choice:

- Explicitly models seasonality, trend, and cyclicity using autoregressive and moving average components [5].
- Uses saturation to stabilise abnormal data to respond to changes in sales trends.
- Provides interpretable parameters for the analysis of seasonal effects.

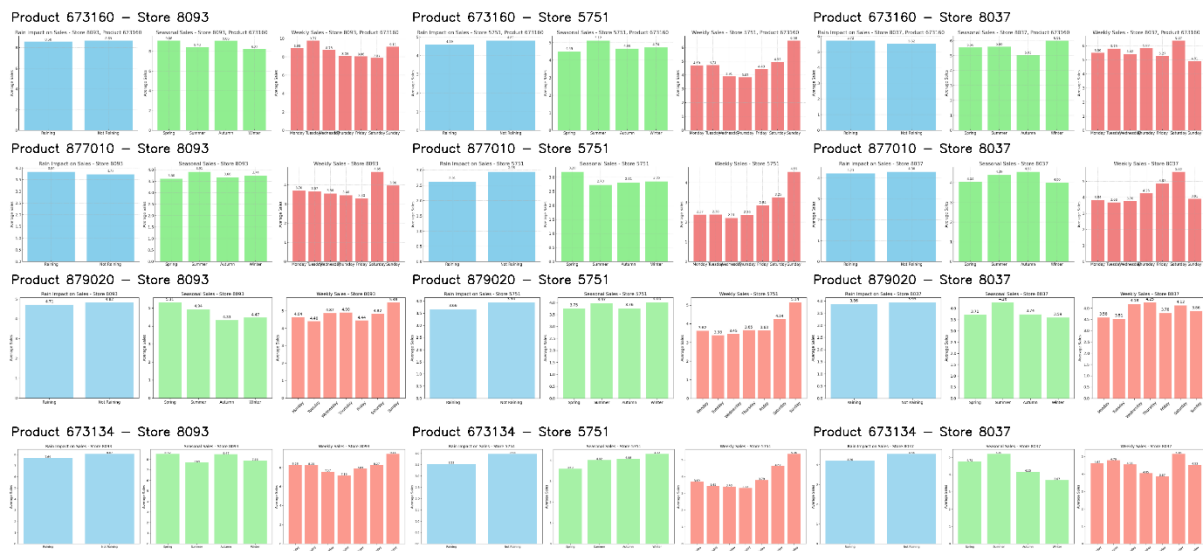
For sales forecasts:

The Seasonal Inverse Moving Average (SARIMA) model is well suited to capture recurring patterns as well as gradually changing trends in sales data and provides a robust statistical basis for time series decomposition.

Model selection will be based on performance after fitting, ensuring the most suitable model for the dataset is chosen. SARIMA and LSTM capture temporal dependencies, with SARIMA emphasising seasonality and LSTM addressing complex sequential patterns. LightGBM and Random Forest leverage structured features to explain variations in sales based on documented external factors. The final model selection will prioritise accuracy, interpretability, and overall effectiveness in capturing sales drivers from available data. This approach ensures that model selection is data-driven and aligns with the specific forecasting requirements of the dataset.

4. Results

The results of analysis into how each of the possible variables that could have been included in the final model affect the daily sales of each of the four products in all three stores are shown in the figure below. The bars represent the daily average sales.



These results show that despite store 5751's daily average sales per day of the week and for rainy days and non-rainy days showing consistent patterns for all of the products, there are no obvious trends between the products within the other stores or between the same products across the different stores. This means that a different model must be fitted for each of the different stores with each product's 'product code' used as a predictor variable so that the model can learn the different sales patterns of each product within the store.

Due to there not being obvious patterns in the variables between the products for each store it is difficult to know which variables should be included in the model and which kind of model to use. To resolve this issue many types of models were fitted (LightGBM, Random Forest, LSTM and SARIMA) with all different variations of variables included in these models to find which one gave the best predictions. The results of the five best models for each of the stores are shown in the tables below.

Store 5751

Model	RMSE	MAE	Features
LightGBM	2.113	1.592	(weekday, season, product code)
LightGBM	2.114	1.579	(weekday, product code)
Random Forest	2.114	1.578	(weekday, product code)
Random Forest	2.125	1.612	(weekday, season, product code)
LightGBM	2.142	1.609	(weather, weekday, season, product code)

Store 8093

Model	RMSE	MAE	Features
Random Forest	3.878	2.813	(weekday, product code)
LightGBM	3.881	2.814	(weekday, product code)
Random Forest	3.929	2.860	(product code)
LightGBM	3.929	2.860	(product code)
LightGBM	3.932	2.862	(rainfall, product code)

Store 8037

Model	RMSE	MAE	Features
Random Forest	2.448	1.840	(rainfall, product code)

LightGBM	2.456	1.808	(season, product code)
Random Forest	2.458	1.810	(season, product code)
LightGBM	2.459	1.843	(product code)
LightGBM	2.459	1.843	(rainfall, product code)

The best model for store 5751 is lightGBM with the days of the week, the seasons and product codes as predictor variables. This is because it gives the lowest Root Mean Squared Error (RMSE). Not including the seasons in this model gives a better Mean Absolute Error (MAE) but RMSE will be prioritised here as it penalises larger errors more than MAE which is necessary when forecasting bakery products due to not wanting to over or under prepare for a day of sales. The model gives a RMSE of 2.113 which means that on average the model's sales prediction deviates by 2.113 per day from the actual sales. Comparing this to the mean sales per day of 9.819 shows that this model gives a reasonable prediction of daily sales but still needs further improvement.

The RMSEs from the best models for stores 8093 and 8037 are 3.878 and 2.448, respectively. For both stores a Random Forest model was found to be best, with the days of the week and product code being included as predictor variables for store 8093, while for store 8037 whether it rained or not on the day and product code were the best predictor variables. The RMSE values can be compared to the daily mean sales of 15.621 for store 8093 and 12.223 for store 8037. These results show a similar story to store 5751's results; that these models provide fairly accurate predictions but would need to be further refined in order to minimise waste.

5. Conclusion

This project successfully developed forecasting models to predict sales of single rolls across 3 stores. The result of these forecasting models is the potential to improve inventory management and reduce waste. The models incorporated historical sales data and external variables such as weather, day of the week, and seasonality. LightGBM emerged as the best-performing model for store 5751 and Random Forest performed best for stores 8093 and 8037, achieving reasonable predictive accuracy. However, further improvements are needed to refine model performance and enhance the ability their ability to predict daily sales.

One key area for improvement is feature selection. While the current models do consider key factors such as weekday, seasons and rainfall, additional variables such as customer footfall, competitor pricing or even local and national events could provide deeper insights into fluctuations in daily demand. Further, more advanced methods of identifying the most relevant predictors such as Recursive Feature Extraction and SHAP values could have been used to eliminate irrelevant noise and improve model efficiency [14].

Additionally, model adaptability is a concern. The models in this study were trained on historical data but do not update as new data becomes available. Implementing online learning techniques, where models continuously update as new sales data is collected, could improve real-time forecasting accuracy. [15] This is particularly relevant in any retail environment, where sales data is continuously collected during opening hours.

Finally, while RMSE was used as the main metric for evaluation, it may not fully capture the effectiveness of the models at forecasting. Utilising other evaluation metrics such as Mean Absolute Scaled Error (MASE) or Weighted RMSE could provide a better assessment of model performance [8].

The developed models offer a solid foundation in predicting sales of small rolls for the three selected stores. However, exploring and implementing models with improved feature selection, improved real-time adaptability and an improved metric for evaluation would significantly improve the accuracy and reliability of the predictions. This would likely result in more effective inventory management, reduced costs, and improved customer satisfaction.

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